Exp No:10. Experiment to understand K-means clustering algorithm for a given data set.

Code:

```
import pandas as pd

df = pd.read_csv(list(uploaded.keys())[0])

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.pairplot(df)
plt.show()

features = df.iloc[:, [3, 4]].values

from sklearn.cluster import KMeans

model = KMeans(n_clusters=5, random_state=42)
model.fit(features)

Final = df.iloc[:, [3, 4]].copy()
Final['label'] = model.predict(features)
```

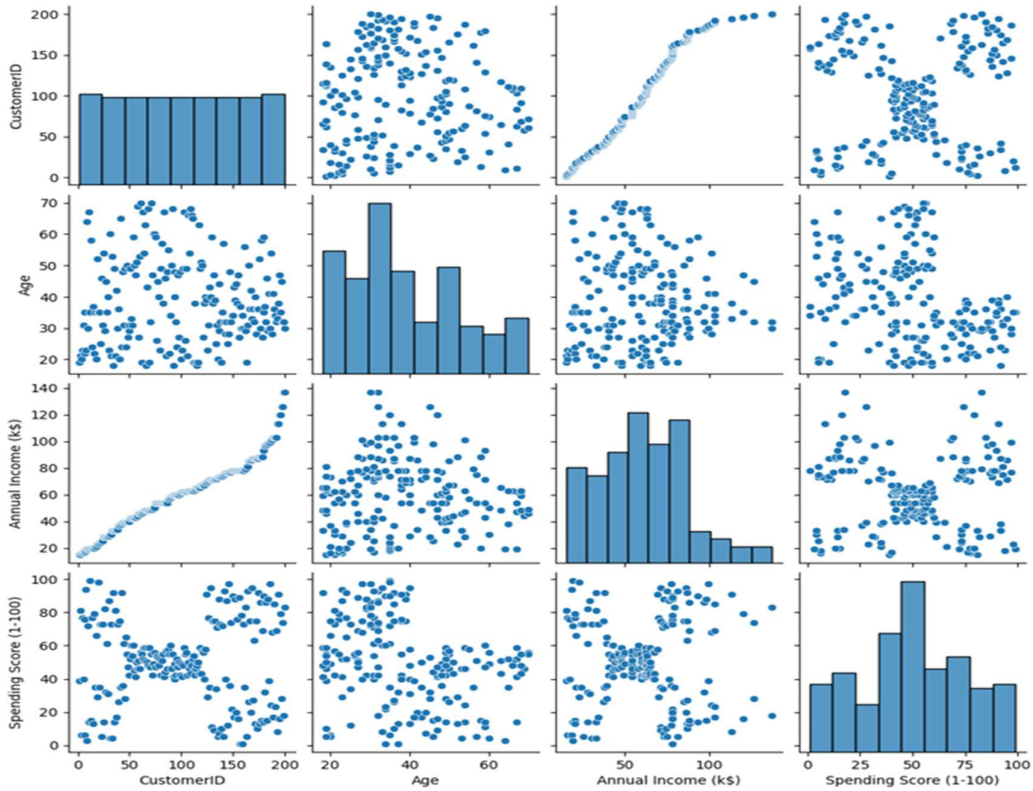
```
Final.head()

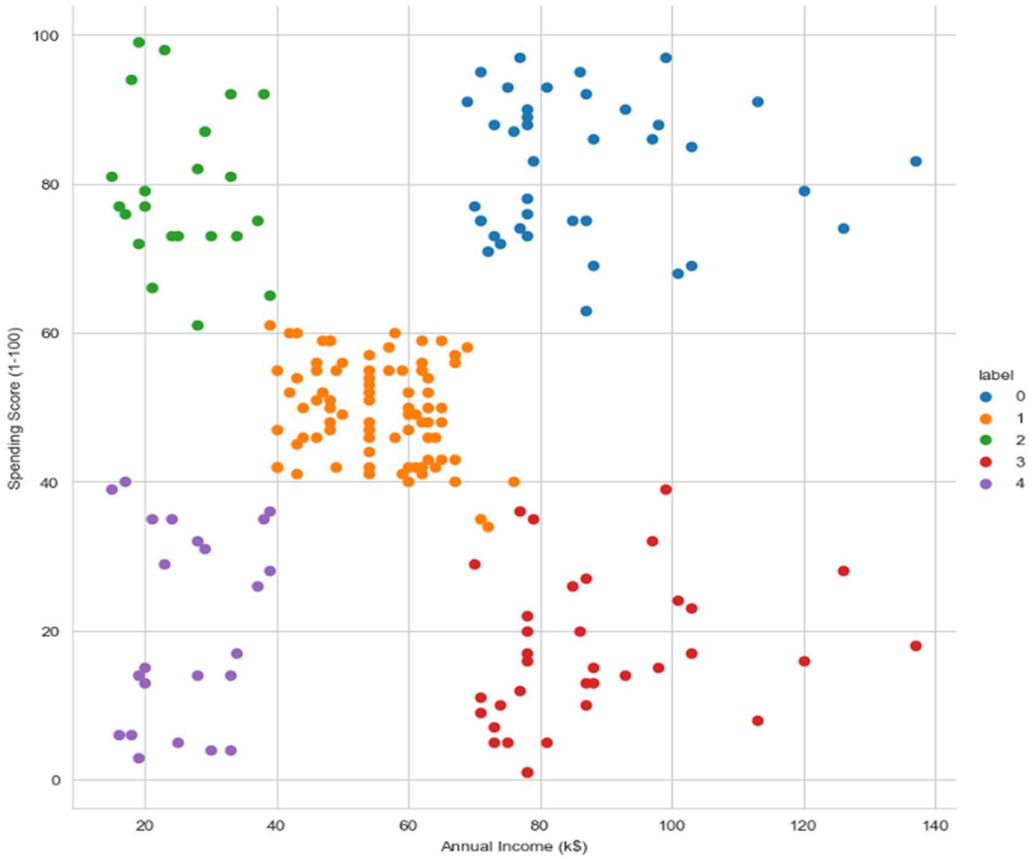

sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8).map(
    plt.scatter, "Annual Income(k$)", "Spending score (1-100)"
).add_legend()

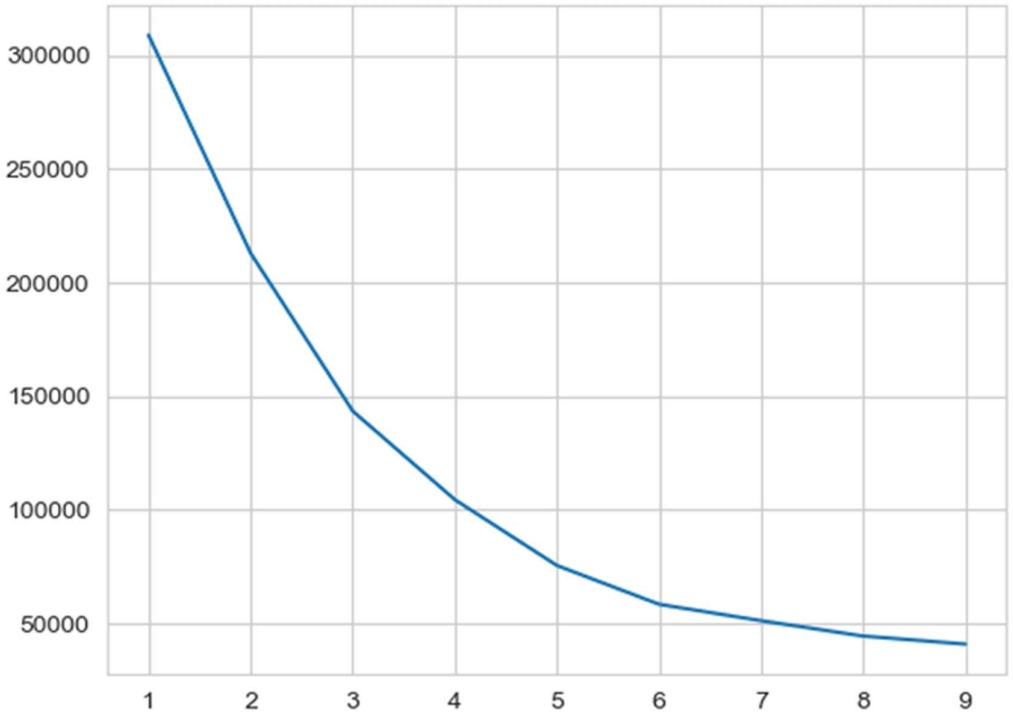
plt.show()
```

Output:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40


CustomerID
Age
Annual Income (k$)
Spending Score (1-100)


Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4



Result:

Thus the code has successfully executed and output is been analysed and it is been verified.